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General Description

The electronic check scale, type ECS, is an independent unit designed to be incorporated in-line in a chain conveyor plant, and is used for semi-automatic or, when check weighing cylinders with the same tare value, for fully automatic check weighing of gas cylinders.

When operating in semi-automatic mode, the tare values of the cylinders are manually encoded by an operator by means of a keyboard on a control terminal When operating in fully automatic mode, the operator encodes one fixed tare value, after which the process runs fully automatically.

The check scale is based on the electronic control/weighing computer type CUC, which is an independent unit that controls and checks the actual check weighing process.

The electronic check scale type ECS consists of the following main components:

- Lower scale with load cell
- Lifting table
- Control terminal on adjustable support with control/weighing computer, keyboard, display and I/O control
- Power supply for connection to main power installation
- Cylinder out-pusher for automatic sorting out of overfilled and underfilled cylinders
- Cabinet with compressed air components and electrical components for connection to electrical power installation
- Air service unit for connection to compressed air installation
- Specially designed chain conveyor section, including necessary automatic chain conveyor controls (stop devices, photocells, etc.)

Scope of Supply

The check scale is supplied complete to be incorporated in a chain conveyor plant (level 550 mm) including corresponding lower scale with load cell, lifting table, control terminal on adjustable support with control/weighing computer, keyboard, display and I/O control, power supply for connection to main power installation, cylinder out-pusher, cabinet with compressed air components and electrical components for connection to electrical power installation, air service unit for connection to compressed air installation, and specially designed chain conveyor section, including necessary automatic chain conveyor controls (stop devices, photocells, etc.).

Additional Equipment

(Only included if specified in quotation)

- Chain conveyor plant
- Sort-out conveyor for overfilled and underfilled cylinders
- Standard spare parts set

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Technical Data

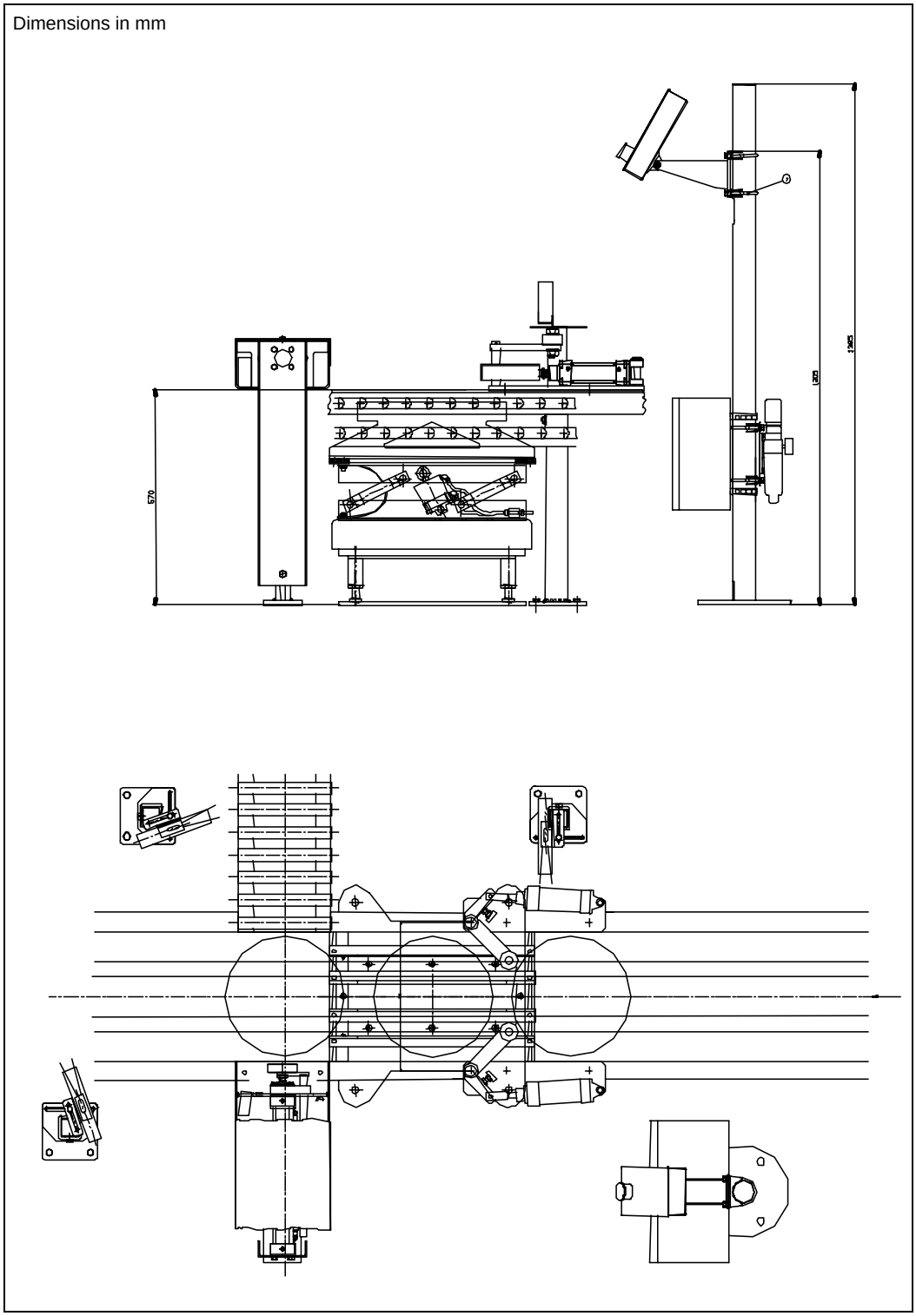
Capacity and conditions	• When operating in semi-automatic mode (manual encoding of tare values): Max. 1300 cylinders/hour, depending on the following factors: – The chain conveyor speed – The quality of the tare value marking on the cylinders – The operator • When operating in fully automatic mode (with fixed tare value): Max. 1500 cylinders/hour, depending on the chain conveyor speed
Cylinder dimensions (requirements and limitations)	• Cylinder diameter: – Min. 220 mm - Max. 410 mm – Max. diameter variation for all cylinders: 50 mm
Necessary connections (specifications and quality requirements)	• Compressed air installation: – Connection size: G1/2" BSP – Working pressure: Min. 0.6 MPa. (6 bar) - Max. 1.0 MPa. (10 bar) – Quality: The quality of the compressed air should be in accordance with ISO 8573.1, class 2-3-2 [1)] • Electrical power installation (via the power supply): – Voltage: 1-phase + neutral + earth (voltage variation from 85 to 264 V AC) – Frequency: 50/60 Hz • Earth net: – Max. transition resistance between earth wire and earth: ≤ 2.0 Ohm
Consumption data	• Compressed air [2)]: – Approx. 5.2 normal litres/cylinder with a sorting out ratio of max. 2%, which corresponds to a max. consumption of 6.240 normal litres/hour at a capacity of 1200 cylinders/hour with 24 sort outs • Electrical power: – < 100 w
Weight	• Approx. 130 kg
Static surface load	• < 0.1 N/mm² (1.0 kp/cm²)
Approvals	• The machine is designed for use in hazardous areas classified as Zone 1 according to IEC 79-10 and Class I, Division 1 according to NEC (USA), article 500 • The weighing computer type CUC and the load cell are approved according to the EU weight standard EN 45501 • The weighing accuracy is according to OIML R76-1, accuracy class III
Other data	• The weighing computer calculates and shows the weight in kilograms according to the SI standard • Weighing range: Min. 5 kg - Max. 120 kg • Display division: 100 g (e.g. 15.1 kg, 15.2 kg, 15.3, etc.) • The check scale is designed for installation in locations in which the temperature may vary between -10°C and +40°C

[1)] The compressed air must not contain particles larger than 1 µm. The pressure dew point of the compressed air at a pressure of 0.7 MPa (7 bar) shall be at least 10°C lower than the lowest ambient temperature to which the compressed air system is exposed at the coldest place of the plant during the year (class 3 ~ -20°C). The maximum oil contents in the compressed air must not exceed 0.1 mg/m³.

[2)] The compressed air consumption can be used for dimensioning the compressor capacity (FAD = Free Air Delivery) based on local conditions. The specified compressed air consumption is based on the following standard reference conditions: An inlet pressure of 0.1 MPa (1 bar), a suction temperature of 20°C and a relative humidity of 60%.

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Dimensions in mm



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